GOVERNMENT OF SINDH

SINDH GOVERNMENT NOTIFICATION ISSUING AUTHORITY (SGNIA)

Comprehensive System Architecture, Technical Stack & Government Administrative Workflow Guide

A Technical and Operational Blueprint detailing Languages, Modules, Hierarchical Secretariat Governance, and Cryptographic Document Verification

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Project Name:	SGNIA (Sindh Govt Notification Issuing Authority)
Live Production URL:	https://sgnia-portal.onrender.com
Public Mirror URL:	https://basitpathan45.github.io/sgnia-portal/
Database Engine:	Supabase PostgreSQL (AWS Cloud Tier)
Administrative Scope:	Government of Sindh Secretariat (43 Active Departments)
Document Type:	Official System Architectural & Operational Blueprint
Target Audience:	System Architects, Evaluators, Secretariat Staff & Auditors
Security Standard:	Ed25519 Asymmetric Signatures & SHA-256 Hash Chain

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1. Executive Summary & Vision

The **Sindh Government Notification Issuing Authority (SGNIA)** is an enterprise digital public infrastructure engineered to modernize, secure, and streamline the lifecycle of official government gazettes, notifications, advisories, and administrative orders across all **43 departments of the Government of Sindh**.

Historically, governmental circulars were distributed via paper memos prone to forgery, manual courier delays, and fraudulent circulation on social media without authentic verification. SGNIA solves this foundational governance challenge by coupling an automated **Secretariat Workflow (Creator-to-Approver hierarchy)** with **Ed25519 asymmetric cryptography** and **SHA-256 tamper-evident integrity hashing**. Every issued document carries a verifiable QR code leading directly to its digital ledger state on the official provincial portal.

2. Technical Stack & Programming Languages

SGNIA is architected as a high-performance web application following multi-tier security standards. Below is the breakdown of each programming language, framework, and tool utilized:

Layer / Domain	Technology / Language	Core Purpose & Operational Role
Frontend Markup	HTML5 Semantic Architecture	Provides accessible, SEO-structured interfaces for both public citizens and authenticated secretariat staff.
Frontend Styling	CSS3 (Variables & Responsive)	Implements Sindh Govt green branding, executive typography, and persistent **Day/Night Mode** via CSS variables.
Client Scripting	Vanilla JavaScript (ES6+)	Powers real-time search filtering across 43 departments, instant client-side SHA-256 PDF hashing, tabs, and modals.
Backend Engine	Python 3.11+	Core server language providing robust security, cryptography bindings, and automated document generation.
Web Framework	Flask 3.0 (Python WSGI)	Orchestrates MVC blueprints (public, staff, auth), session security, and RBAC.
Relational DB	Supabase PostgreSQL (Cloud)	Enterprise cloud database storing departments, staff credentials, notifications, keys, and immutable audit logs.
ORM & Data Layer	SQLAlchemy 2.0	Translates relational tables into Python models, manages foreign keys, and prevents SQL injection attacks.
Cryptography	Cryptography (Ed25519 & SHA-256)	Generates asymmetric signing keys, executes digital signatures, and creates mathematical anti-tamper fingerprints.
Document Engine	PyMuPDF (Fitz) & Custom Canvas	Generates official Sindh Govt notification PDFs with header emblem, typography, and embedded verification QR codes.
OCR & Digitization	PyPDF, PyMuPDF & Tesseract	Extracts textual contents from legacy and uploaded PDF documents directly on the server to auto-fill draft summaries.
Cloud Deployment	Render PaaS & Gunicorn WSGI	24/7 cloud hosting running multi-threaded Gunicorn workers with TLS/SSL encryption and continuous deployment.

3. Core System Features & Citizen Value

• **Public Document Archive & Registry:** Enables citizens, journalists, and agencies to search, filter, and inspect published orders across all 43 provincial departments with real-time dynamic counters.

• **Instant Cryptographic Document Verification:** Allows anyone holding a printed or digital PDF to verify authenticity by entering its Reference or uploading the PDF. The browser hashes the file locally and checks it against the database.

• **Smartphone QR Code Verification Guide:** Every auto-generated document contains a secure QR code. Scanning it with any mobile camera immediately opens the official portal verification page showing live integrity status.

• **Automated Official PDF Generation Engine:** When a Creator leaves upload blank, SGNIA automatically builds an official Sindh Government notification PDF with official fonts, emblem, references, date stamps, and QR code.

• **Multi-Role Staff Directory with Tabbed Categorization:** Segregates personnel into distinct tabs (Super Admin, Approver, Creator, Auditor) with dedicated search bars, name editing, password reset modals (min 12 chars), and safe deactivation/deletion.

• **OCR & Text Digitization Suite:** Instantly extracts embedded text from existing digital circulars and scanned archives, populating draft subject and body fields without manual re-typing.

• **Tamper-Evident SHA-256 Audit Trail:** Records every administrative action in an immutable hash-chained ledger to ensure strict anti-corruption accountability.

• **Day / Night Mode Experience:** Provides a persistent visual theme switcher in the topbar, saving user preferences in local storage while reducing eye fatigue during nighttime administrative duties.

4. Sindh Secretariat Hierarchy & Administrative Structure

In the parliamentary administrative system of the Government of Sindh, governance follows the **Rules of Business**. Below is the statutory hierarchy and how it maps to SGNIA portal roles:

Secretariat Rank	SGNIA Portal Role	Statutory Authority & Administrative Duty
Chief Secretary Sindh (CS) (Apex Administrative Head)	Super Admin / System Executive	Exercises overall executive oversight of provincial governance; reviews systemic audit trails and cross-departmental coordination.
Secretary / ACS (Principal Accounting Officer)	Approver (Competent Authority)	Holds ultimate statutory authority to approve and issue gazette notifications on behalf of the Governor and Government of Sindh. In SGNIA, clicking 'Approve & Publish' cryptographically seals the notification.
Special / Additional Secretary (Senior Wing In-Charge)	Approver / Co-Signatory	Reviews summaries, examines legal/financial implications, and approves subordinate notifications delegated by the Secretary.
Deputy Secretary (DS) (Branch Supervisor)	Senior Reviewer / Approver	Supervises the clerical and executive workflow of specific departmental wings; scrutinizes drafts submitted by Section Officers.
Section Officer (SO) / ASO (Branch Executive Officer)	Creator (Draft Originator)	Initiates draft orders, circulars, and notes. Enters document dates, official references, and text in SGNIA, generating the draft PDF for review.
Director General Audit / SGA&CD Oversight	Auditor (Independent Monitor)	Possesses unrestricted view-only access across all records, version histories, and the tamper-evident cryptographic audit log.

5. Step-by-Step Notification Lifecycle & Workflow

The workflow within SGNIA strictly mirrors the official file movement (Rules of Business) of the Sindh Secretariat, augmented with modern cryptographic validation. Below is the operational sequence:

Phase 1: Inception & Drafting (Section Officer / Creator)

A department Section Officer logs into SGNIA and navigates to 'Create Notification'. They select their Department (e.g., Home Department), category (Executive Order, Notification, Advisory), Subject, and Body text. The system automatically builds the official Sindh Government formatted PDF with embedded emblem and integrity hash, storing it in 'Draft' state.

Phase 2: Submission for Review

The Creator verifies the generated PDF document. Once satisfied, they click 'Submit for Review'. The notification transitions status from Draft to Pending Approval and immediately appears in the Approver's departmental queue.

Phase 3: Administrative Scrutiny & Auto-Publish (Competent Authority / Approver)

The Secretary or authorized Additional Secretary accesses 'Approvals & History'. They review the full text and inspect the generated PDF. Clicking Approve & Publish seals the notification into law, signs its SHA-256 fingerprint with the Ed25519 key, and broadcasts it immediately to the Public Portal ledger.

Phase 4: Public Dissemination & Citizen Verification

The gazette notification is live on sgnia-portal.onrender.com. Citizens, law enforcement, and courts verify the printed paper copy by scanning its QR code or uploading the PDF to check its cryptographic match.

Phase 5: Amendments & Corrigendum (Version Control)

If the Government subsequently amends dates or clauses, the original Version 1 (v1) remains permanently archived to preserve legal evidence. A new Version 2 (v2) is drafted, approved, and published, establishing an unbroken chain of statutory history.

6. Security Architecture & Anti-Tamper Guarantees

Security Layer	Mechanism	Threat Protection & Integrity Guarantee
Document Integrity	SHA-256 Hash Digest	Every byte of the PDF is cryptographically hashed. If even a single character is altered on a forged copy, the hash completely diverges, failing verification.
Statutory Authenticity	Ed25519 Elliptic Curve Signatures	Published notifications are mathematically signed using Ed25519 cryptography, providing non-repudiation that only the Government of Sindh issued this document.
Audit Ledger Chain	Tamper-Evident Hash Chain	Each entry in the Audit Log includes the SHA-256 hash of the previous log entry. Any attempt to alter historical logs corrupts the chain, alerting authorities immediately.
Access Control (RBAC)	Departmental Scoping	Creators cannot approve drafts; Approvers cannot approve outside their department; only Super Admins manage accounts and signing keys.
Credential Safety	Salted Hashes (Werkzeug)	Passwords are never stored in plaintext and require a minimum of 12 characters, protecting against brute-force attacks.

Conclusion

The SGNIA platform establishes a new benchmark for provincial e-governance in Pakistan. By integrating modern Python/PostgreSQL architecture with cryptographic verifiability and the official Rules of Business of the Sindh Secretariat, SGNIA ensures that government orders are authentic, instantaneous, legally binding, and resilient against digital forgery.